Design and Control of Heterogeneous Azeotropic Column System for the Separation of Pyridine and Water

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The design and control of two types of separation systems using heterogeneous azeotropic distillation have been studied in the literature. One type (e.g., isopropyl alcohol dehydration) is to add a light entrainer (cyclohexane) into the system so that a minimum-boiling ternary azeotrope is formed which can split into two liquid phases in a decanter. This type of system also introduces two additional azeotropes (isopropyl alcohol−cyclohexane and water−cyclohexane), thus dividing the ternary system into three distillation regions. Another type (e.g., acetic acid dehydration) does not contain azeotrope in the original system. However, due to a tangent pinch near the pure water end an entrainer (isobutyl acetate) is added into the system to aid the separation. This type of system has only one binary heterogeneous azeotrope (water−isobutyl acetate); thus there is only one distillation region. In this paper, design and control of a different residue curve map (RCM) type of separation system utilizing heterogeneous azeotropic distillation will be studied. An example of this RCM type is pyridine and water separation using toluene as entrainer. Adding toluene into this system introduces two additional azeotropes, one is minimum-boiling binary heterogeneous azeotrope (water−toluene) and the other one is a binary homogeneous azeotrope (pyridine−toluene). There is no ternary azeotrope for this system. Two alternative design flow sheets are compared in this paper to find the one which is most economical and also meet stringent product purity specifications. A simple overall control strategy of this process has also been developed which requires only one temperature control loop in each column.

1. Introduction

Heterogeneous azeotropic distillation is commonly used in industry to separate mixtures of close relative volatility and breaking azeotropes. The advantage of this separation is to utilize a natural liquid−liquid separation in a decanter. Widely used industrial applications include organic alcohol and acetic acid dehydration systems. A nice review paper by Widagdo and Seider[1] showed that parametric sensitivity, multiple steady states, and long transient and nonlinear dynamics were found by many authors. These heterogeneous azeotropic distillation columns are known to be difficult to operate and control.

Ethanol and isopropyl alcohol dehydration are the two most commonly seen industrial examples for alcohol dehydration. For the ethanol dehydration, Kovach and Seider[2] performed step tests on feed rate and operating variables for an industrial column for dehydrating ethanol using di-*sec*-butyl ether (DSBE) as entrainer and found erratic behavior which was attributed to parametric sensitivity. Bozenhardt[3] proposed a control strategy involving average temperature control, online break point position control, and five feed-forward control loops for the ethanol + ether + water system. Rovaglio et al.[4−6] proposed average temperature control and two feed-forward control loops for the ethanol + benzene + water system. Müller et al.[7] were able to fit data obtained by a laboratory tray column with an equilibrium stage model for an ethanol + cyclohexane + water system, and later Müller and Marquardt[8] experimentally reported evidence of multiple steady states of this system. Luyben[9] proposed a plantwide control strategy for an ethanol + benzene + water three-column system showing counterintuitive behavior in the aqueous level control loop.

For the isopropyl alcohol (IPA) dehydration, previous studies[10−13] of this system by using cyclohexane (CyH) as an entrainer showed that the optimum operation point should be located at a critical reflux, a transition point at which the distillation path switches from a route that passes through IPA + H_2O azeotrope to one that passes through IPA + CyH azeotrope. At this critical reflux, a very high purity IPA product can be obtained with minimum energy consumption and maximum product recovery. However, from bifurcation analysis, this critical reflux point is right at the edge of the top stable branch and the middle unstable branch of steady-state operating conditions; thus this steady state is extremely sensitive to any small perturbations. An inverse double loop control strategy was recommended to maintain a steady column temperature profile. This control strategy was verified via dynamic simulation and experimental test. Chien et al.[14] extended the previous study of a single heterogeneous azeotropic distillation column to include a recovery column. Arifin and Chien[15] further extended the study for more diluted fresh feed and proposed a two-column system with one column serving as the preconcentrator column as well as the recovery column.

For acetic acid (HAc) dehydration system, Pham and Doherty[16] listed in a table examples of heterogeneous azeotropic separations using ethyl acetate, *n*-propyl acetate, or *n*-butyl acetate as entrainer. Siirola[17] used acetic acid dehydration as an example to demonstrate a systematic process synthesis technique for the conceptual design of a process flow sheet. Ethyl acetate was used as the entrainer in that paper to design a complete acetic acid dehydration process with multiple-effect azeotropic distillation and heat integration. Wasylkiewicz et al.[18] proposed using a geometric method for the optimum design of a HAc dehydrating column with *n*-butyl acetate as the entrainer. Chien et al.[19] studied the design and control of a HAc dedydration system via heterogeneous azeotropic distillation using isobutyl acetate as entrainer. Several other papers[20−22] addressed the design and control of acetic acid dehydration process with *p*-xylene or *m*-xylene feed impurity. A side stream

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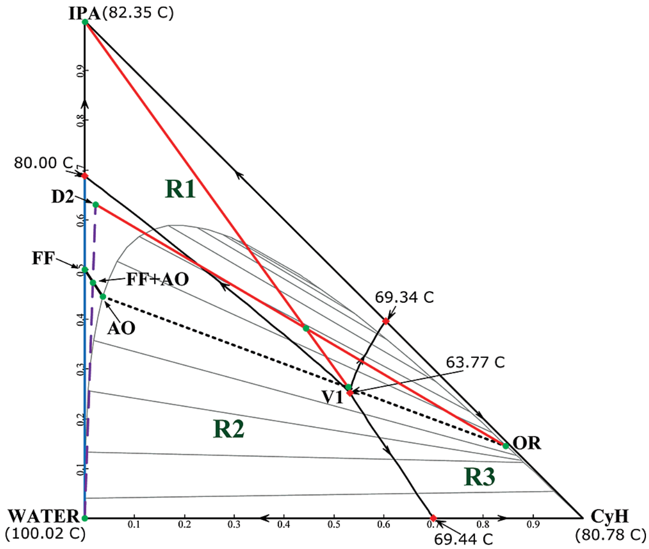

Figure 1. RCM, liquid−liquid equilibrium (LLE), and material balance lines for isopropyl alcohol dehydration system.

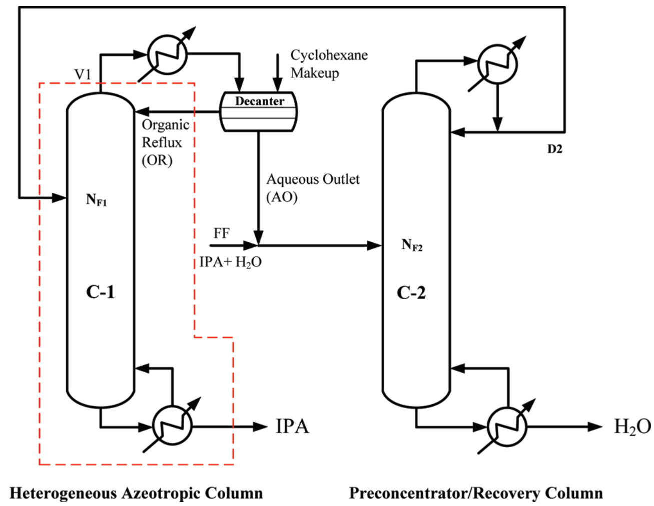

Figure 2. Design flow sheet for isopropyl alcohol dehydration system.

is needed in the design to handle the accumulation problem of this impurity.

In this paper, a heterogeneous azeotropic column system with different residue curve map (RCM) topology is investigated. The studied system is to separate pyridine and water using toluene as entrainer. This separation system was studied by Pommier et al.[23] for the operating strategy via heterogeneous batch distillation. This RCM topology belongs to the Serafimov's class 3.0-2[24] (122 by Matsuyama and Nishimura[25] classification). The binary azeotrope of toluene and water has the minimum temperature of the ternary system and also is inside the liquid−liquid boundary. The design and control of this RCM type will be investigated in the following sections.

2. Conceptual Design of Various RCM Types

We will start this paper by explaining how the heterogeneous azeotropic distillation works in separating a close-boiling mixture or a mixture with an azeotrope. Four systems with different RCM topology will be used as examples to illustrate the column sequence for the separation.

2.1. Isopropyl Alcohol Dehydration Using Cyclohexane as Entrainer (cf. Arifin and Chien[15]). As shown in Figure 1, the IPA−water mixture has an azeotrope with composition of around 31 mol % H_2O and azeotropic temperature of 80 °C. By adding cyclohexane (CyH) into the system, as shown in this figure, three additional azeotropes are formed. One desirable ternary azeotrope is heterogeneous with an azeotropic temperature of 63.77 °C that is the minimum temperature for the entire ternary system. Another two azeotropes (one heterogeneous and another one homogeneous) all have higher azeotropic temperatures. As shown in Figure 1, three distillation boundaries divide the RCM into three distillation regions with the stable nodes of regions R1, R2, and R3 to be IPA, water, and CyH, respectively. The feasible column sequence shown in Figure 2 was suggested by Ryan and Doherty[26] for their ethanol dehydration system and then used by Arifin and Chien[15] for this separation system. The design flow sheet contains a combined column that serves as a preconcentrator column and also as a recovery column and another heterogeneous azeotropic distillation column.

One purpose of the combined preconcentrator/recovery column is to remove part of the water in the fresh feed (assume the fresh feed composition is 50 mol % IPA to agree with a true application of a waste IPA stream in the semiconductor industry). The other purpose is to recover significant amounts of the IPA product in the aqueous outlet stream. The bottom product of this combined column will be water, while the distillate composition of this combined preconcentrator/recovery column will be a ternary mixture but close to the azeotropic composition of IPA−water. Note that this column is operated inside the R2 distillation region.

This distillate stream is fed into a heterogeneous azeotropic column. With the aid of adding another component (cyclohexane) into the system through an organic reflux stream, the bottom composition will approach pure IPA composition and the column top vapor will approach the lowest temperature of the ternary system, which is the ternary azeotrope. Note that this column is operated inside the R1 distillation region. This azeotrope is heterogeneous, so by condensing and cooling the top vapor and feeding the condensate at 40 °C to a decanter, natural liquid−liquid separation into an organic phase and an aqueous phase will occur. The organic phase, which is rich in entrainer, is designed to go back to the heterogeneous azeotropic column. The composition of the aqueous phase still contains significant amounts of the IPA product, so it is fed into the combined column. The material balance lines in Figure 1 clearly explain the conceptual design of this separation flow sheet in Figure 2.

Another point worth mentioning is that the products of the column sequence (IPA and water) are in different distillation regions (see Figure 1). This normally cannot happen for a column sequence with relatively linear distillation boundaries, as is the case for this system. The reason that the distillation boundary can be crossed in this system is because of the occurrence of liquid−liquid separation in the decanter.

2.2. Acetic Acid Dehydration Using Isobutyl Acetate as Entrainer (cf. Chien et al.[19]). As can be seen in Figure 3, the acetic acid (HAc) and water do not form an azeotrope. However, it is well-known that this system exhibits a tangent pinch near the pure water end. Thus, using simple distillation would require many stages to obtain high-purity products. A more economical way for this separation system is to add entrainer (e.g., isobutyl acetate, iBuAc) and operate it as a heterogeneous azeotropic distillation column. Adding entrainer introduces one azeotrope in the ternary system with composition of around 64 mol % H_2O and azeotropic temperature of 87.72 °C. This azeotropic temperature is the lowest temperature of the ternary system; thus, would come out of the heterogeneous azeotropic distillation

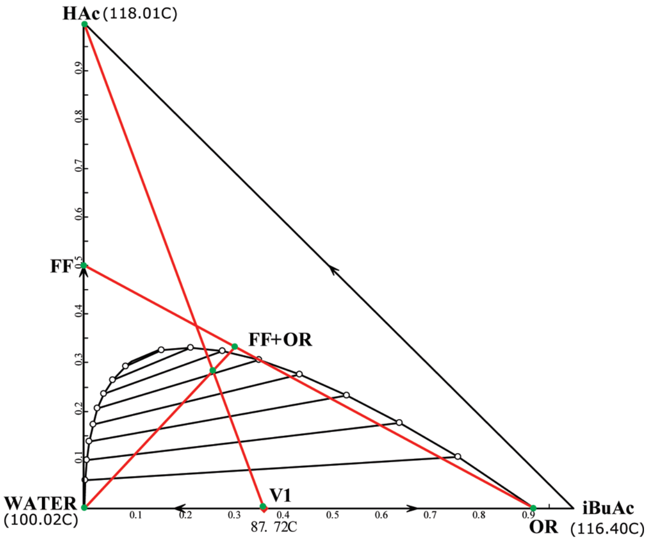

Figure 3. RCM, LLE, and material balance lines for acetic acid dehydration system.

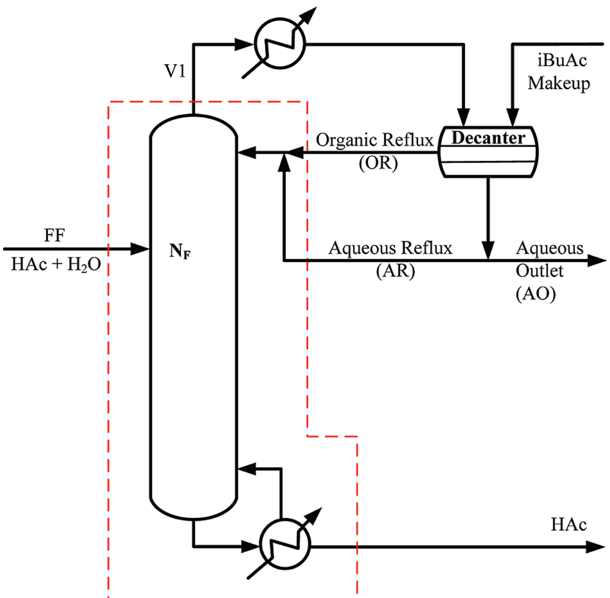

Figure 4. Design flow sheet for acetic acid dehydration system.

column as the top vapor, while acetic acid having the highest temperature will be the bottom product.

This azeotrope is heterogeneous, so by condensing and cooling the top vapor and feeding the condensate at 40 °C to a decanter, natural liquid−liquid separation into an organic phase and an aqueous phase will occur. The organic phase, which is rich in entrainer, is designed to go back to the heterogeneous azeotropic column while the aqueous composition is near pure water and thus can be drawn out of the system. A small entrainer makeup stream is needed to feed into the system at the decanter to balance the small entrainer loss mainly in the aqueous outlet stream.

The unique feather of this separation system is that part of the aqueous phase should be recycled back to the heterogeneous azeotropic column to let the top vapor composition stay very close to the azeotrope, thus preventing loss of acetic acid in the aqueous outlet stream. The material balance lines in Figure 3 clearly explain the conceptual design of this one-column flow sheet in Figure 4.

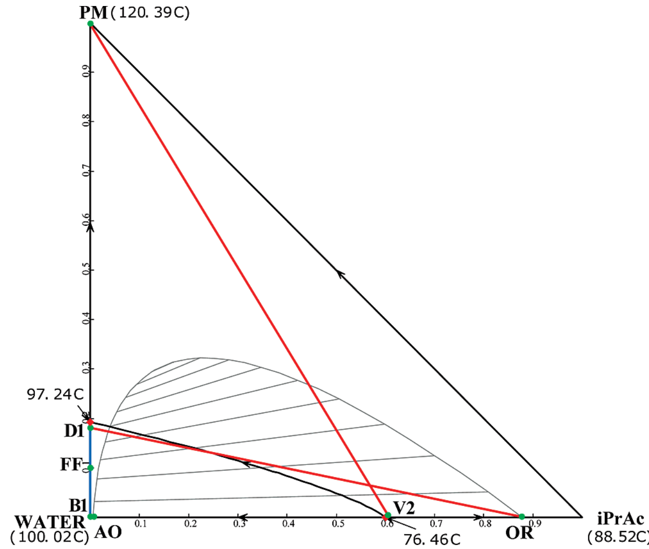

Figure 5. RCM, LLE, and material balance lines for PM dehydration system.

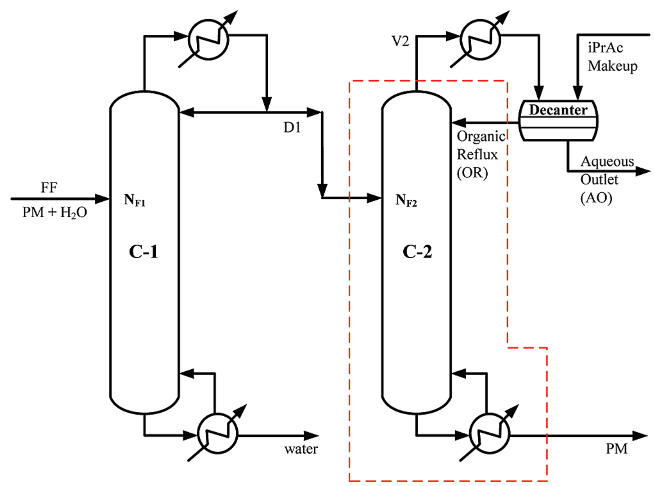

Figure 6. Design flow sheet for PM dehydration system.

2.3. Propylene Glycol Monomethyl Ether−Water Separation Using Isopropyl Acetate as Entrainer. An industrially relevant separation problem is used here as a third example. As shown in Figure 5, the propylene glycol monomethyl ether−water (PM−water) mixture has an azeotrope with composition around 81 mol % H_2O and azeotropic temperature of 97.24 °C. By adding isopropyl acetate (iPrAc) into the system, an additional azeotrope is formed that has a minimum azeotropic temperature of 76.46 °C. This azeotrope is also inside the liquid−liquid equilibrium (LLE) envelope. With the aid of the information in this figure, the feasible column sequence can be designed. As shown in Figure 6, the flow sheet features a preconcentrator column and a heterogeneous azeotropic distillation column.

The purpose of the preconcentrator column is to remove part of the water in the fresh feed (assume the fresh feed composition is 90 mol % water). The distillate composition of this preconcentrator column will approach the azeotropic composition of PM−water. This distillate stream is fed into a heterogeneous azeotropic column. With the aid of adding another component (iPrAc) into the system through an organic reflux stream, the bottom composition will approach pure PM composition and the column top vapor will approach the lowest temperature of

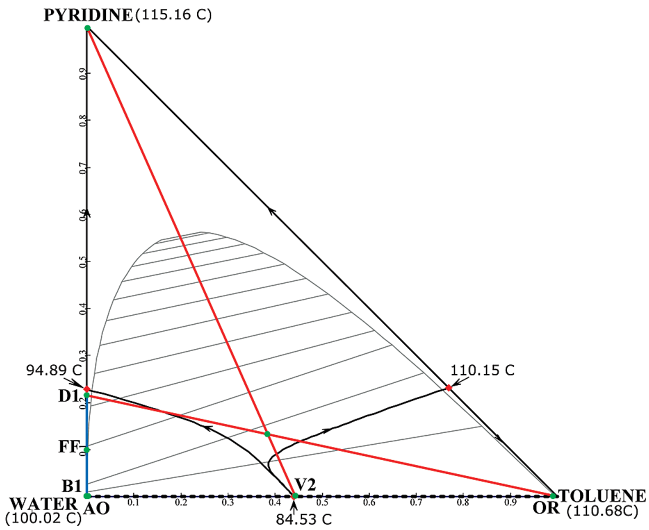

Figure 7. RCM, LLE, and material balance lines for pyridine dehydration system with column in sequence.

the ternary system, which is the azeotropic composition of isopropyl acetate−water.

This azeotrope is heterogeneous. Thus, by condensing and cooling the top vapor to 40 °C and feeding the condensate to a decanter, natural liquid−liquid separation into an organic phase and an aqueous phase will occur. The organic phase, which is rich in entrainer, is recycled back to the heterogeneous azeotropic column. The composition of the aqueous phase is pure enough that it can be drawn out of the system as aqueous product. A small entrainer makeup stream is needed to feed into the system at the decanter to balance the small entrainer loss mainly in the aqueous outlet stream. The material balance lines for each of the two columns (shown in Figure 5) clearly explain the conceptual design of this separation flow sheet.

2.4. Pyridine−Water Separation Using Toluene as Entrainer (cf. Pommier et al.[23]). As shown in Figure 7, a pyridine−water mixture has an azeotrope with a composition of around 77 mol % H_2O and azeotropic temperature of 94.89 °C. By adding toluene into the system, two additional azeotropes are formed. One desirable azeotrope (toluene−water) is heterogeneous with an azeotropic temperature of 84.53 °C, which is the minimum temperature for the entire ternary system. Another azeotrope between pyridine and toluene is also formed with a higher azeotropic temperature of 110.15 °C. As shown in Figure 7, the distillation boundaries divide the RCM of the ternary system into three distillation regions. The feasible column sequence is first thought to be as in the previous system containing a preconcentrator column and a heterogeneous azeotropic distillation column (see Figure 8).

The purpose of the preconcentrator column is again to remove part of the water in the fresh feed (assume the fresh feed composition is 90 mol % water). The lowest temperature in this ternary system is the toluene−water azeotrope, and the highest temperature is pyridine. Therefore the top vapor of the heterogeneous azeotropic column is designed to approach a toluene−water azeotrope and the bottom product is pure pyridine. The material balance lines of the column sequence can also be seen in Figure 7.

One difficulty in the design flow sheet in Figure 8 is that reaching the toluene−water azeotrope requires going through a very narrow funnel (see Figure 7). Hence, this heterogeneous column requires a large number of total stages or is even practically impossible to reach this point (as can be shown in the rigorous simulation in the next section). Failure to reach right at this toluene−heterogeneous azeotrope will result in some pyridine loss from the aqueous outlet stream. An alternative design flow sheet will be proposed in this paper that is more economical and also enhances the operability of the separation system.

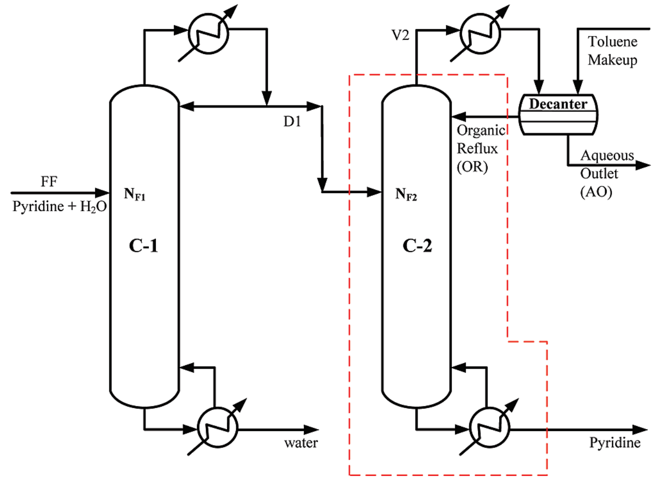

Figure 8. Design flow sheet with column in sequence.

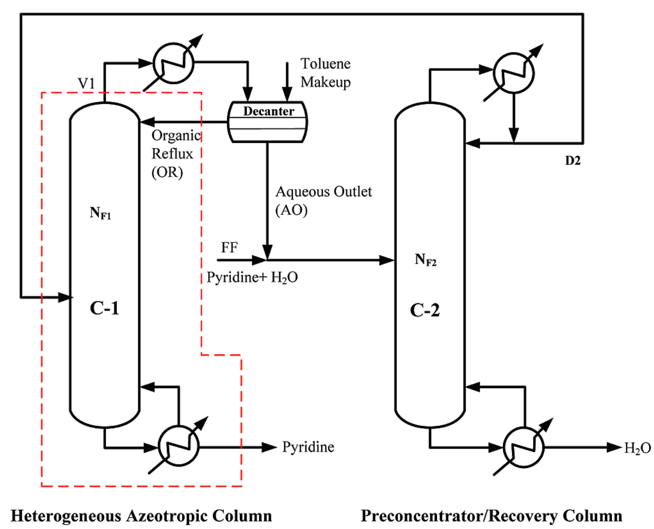

Figure 9. Proposed design flow sheet for pyridine dehydration system.

Although theoretically a preconcentrator column followed by a heterogeneous azeotropic column can do the jobs for this RCM type, however, the worry is if the top vapor is feasible to reach right at this azeotrope. To improve the above design, an alternative two-column system making use of the combined column concept is proposed (see Figure 9) with the material balance lines of the conceptual design shown in Figure 10. Note that the top vapor of the heterogeneous azeotropic column does not need to be strictly placed at the toluene−water azeotrope. Column C-2 serves as a preconcentrator column and also as a recovery column to obtain pure water product. The comparison of this proposed design versus the one in Figure 8 will be shown using a rigorous process simulation in the next section.

For all the analysis above using the information of the residue curve maps and the LLE boundary at 40 °C, the validity of the thermodynamic model to predict correctly the vapor−liquid equilibrium (VLE), LLE, and the azeotropic information is very important. Good sources of the experimental data of VLE, LLE, and azeotropic information can be found in DECHEMA (Gmehling and Onken,[27] Sørensen and Arlt[28]) and in Gme-

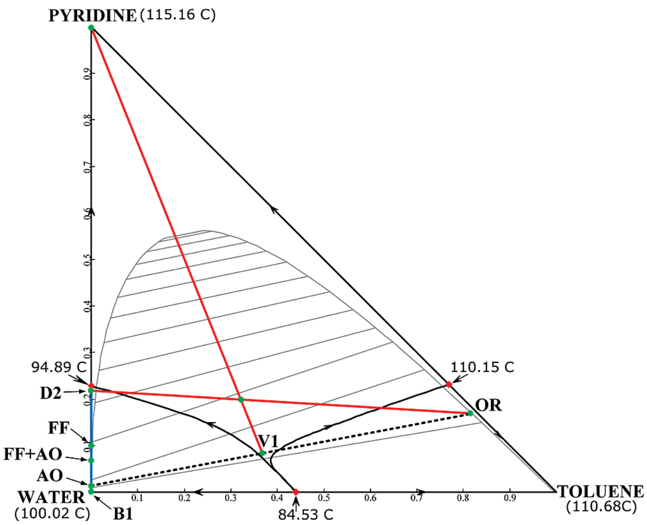

Figure 10. RCM, LLE, and material balance lines for pyridine dehydration system with proposed design flow sheet.

hling.[29] Table 1 shows the NRTL parameters used in the above four systems. The parameter set for the isopropyl alcohol dehydration system is from Wang et al.,[10] and that of the acetic acid dehydration system is from Chien et al.[19] The parameter set for the PM−water separation system is from an industrial partner, and the parameter set for the pyridine−water separation system is from Pommier et al.[23] and Aspen built-in. The vapor phase is assumed to be ideal except for the acetic acid system. In that system, the Hayden and O'Connell[30] model was used to account for the association in the vapor phase.

3. Design Flow Sheet of the Pyridine Water Separation System

3.1. Flow Sheet of the Proposed Design. The optimal process flow sheet of the proposed design in Figure 9 will be obtained first by minimization of the total annual cost (TAC) with five design variables. There are the following: the pyridine distillate composition (XD2) of the combined column, the total number of stages for the heterogeneous azeotropic column and the combined column (N_1 and N_2), and the two feed stage locations (N_{F1} and N_{F2}). The product specification of pyridine at the bottom of the heterogeneous azeotropic column is set to be 99.9 mol %. The product specification of water at the bottoms of the combined column is also set to be 99.9 mol %. In each simulation run, the pyridine product specification is achieved by varying the reboiler duty of the heterogeneous azeotropic column, and the water product specification is achieved by varying the reboiler duty of the recovery column. The entrainer makeup flow rate will be very small to balance the entrainer loss from the two bottom streams. The flow rate of this stream is also determined to lower the reboiler duty.

The TAC of the overall system includes the annualized capital costs and the operating costs. The annualized capital costs include the following: column shells of the two columns, internal trays of the two columns, and also the reboilers and condensers of the two columns. The payback period was assumed to be 3 years in the calculation. The equations for the cost calculations can be seen from Appendix E of the Douglas[31] process design book. The operating costs include the following: the steam and cooling water required for operating these two columns and also very small entrainer makeup cost.

Sequential iterative procedure was used to obtain the optimal design flow sheet with the procedure summarized in Figure 11.

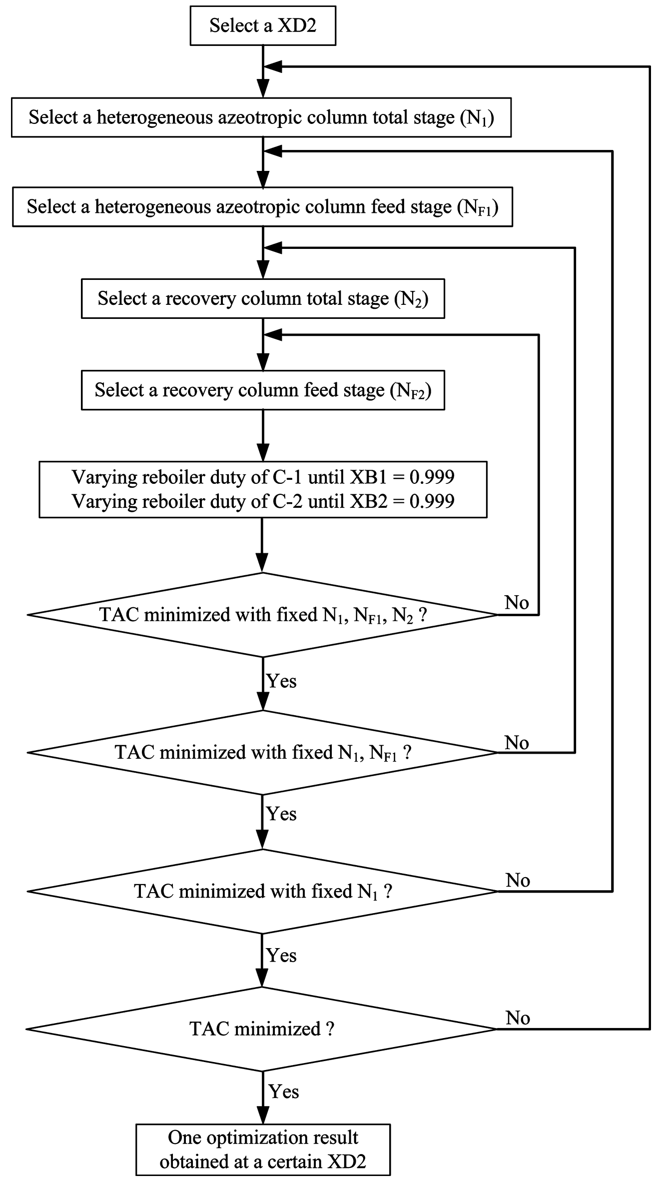

Figure 11. Sequential iterative optimization procedure for this system.

The optimization method is intuitive; however, it is rather time-consuming when the numbers of design variables become large in a plantwide optimization. Figure 12 shows the summary of the TAC plots at each XD2. The optimal distillate composition of the combined column is at 0.210. This pyridine distillate composition represents a trade-off in the overall process design. The costs associated with the combined column will be increased when this distillate composition is purer in pyridine. However, setting XD2 purer in pyridine will make the material balance line connecting points D2 and organic reflux (see Figure 10) further away from V1, thus reducing the recycle flow rate in the overall system. At this distillate composition of 0.210, the TAC of the overall system is minimized. The inner iterative optimization loops found the total stages of the heterogeneous azeotropic column to be 14 with the sixth stage as the feed stage and the total stages of the combined column to be 10 with the sixth stage as the feed stage as well. The optimal design and operating condition can be seen in Figure 13.

3.2. Design Flow Sheet with Preconcentrator Column and Heterogeneous Azeotropic Column in Sequence. From the RCM in Figure 7, the location of V1 is theoretically achievable because it is in the same distillation region with the

Table 1. NRTL Model Parameters for the Studied Systems[a]

	for given component i, component j combination											
	IPA, water	IPA, CyH	CyH, water	HAc, water	HAc, iBuAc	iBuAc, water	PM, water	PM, iPrAc	iPrAc, water	pyridine, water	pyridine, toluene	toluene, water
a_{ij}							−1.530	0	0	0	0	−247.879
a_{ji}							7.626	0	0	0	0	627.053
b_{ij}(K)	185.4	294.6	1629	−211.310	90.268	489.609	431.05	−60.437	186.759	209.420	−30.365	14759.8
b_{ji}(K)	777.3	622.5	2328	652.995	194.416	1809.079	−1998.746	338.092	1483.132	895.315	133.173	−27269.4
c_{ij}	0.5	0.5	0.242	0.3	0.3	0.2505	0.3	0.3	0.2	0.6932	0.2992	0.2
e_{ij}										0	0	35.582
e_{ji}										0	0	−92.718

[a] Aspen Plus NRTL:

$$\ln \gamma_i = \frac{\sum_j x_j \tau_{ji} G_{ji}}{\sum_k x_k G_{ki}} + \sum_j \frac{x_j G_{ij}}{\sum_k x_k G_{kj}} \left[\tau_{ij} - \frac{\sum_m x_m \tau_{mj} G_{mj}}{\sum_k x_k G_{kj}} \right]$$

where

$$G_{ij} = \exp(-\alpha_{ij}\tau_{ij})$$

$$\tau_{ij} = a_{ij} + \frac{b_{ij}}{T} + e_{ij} \ln T$$

$$\alpha_{ij} = c_{ij}, \quad \tau_{ii} = 0, \quad G_{ii} = 1$$

bottom product; however, to reach this location, the composition profile needs to go through a very narrow funnel. In this section, rigorous process simulation is conducted to confirm this observation. Figure 14 shows a simulation result with the total stages of the heterogeneous azeotropic column set at 120. Even with this practically impossible tall column, the pyridine loss through the aqueous outlet stream is still higher than the one in the proposed design flow sheet (0.12 mol % pyridine loss through aqueous outlet stream in this figure vs 0.1 mol % pyridine loss through the bottoms of the combined column in Figure 13). The key difference is that the proposed flow sheet contains a "recovery column" while for this design the aqueous outlet stream goes directly out of the system.

A reasonable specification of pyridine loss through the aqueous outlet stream is set at 0.6 mol %. This specification can be obtained by varying the makeup flow rate. A similar sequential iterative optimization procedure is followed to find the values of the design variables: distillate composition of the preconcentrator column, total stages of the preconcentrator column, feed stage of the preconcentrator column, total stages of the heterogeneous azeotropic column, and feed stage of the heterogeneous azeotropic column. The resulting optimized design flow sheet is displayed in Figure 15 with a TAC comparison summarized in Table 2. The TAC of the proposed design results in slightly smaller TAC than that of the design in this section. The main reason for selecting the proposed design is that the pyridine loss through the water stream is much smaller

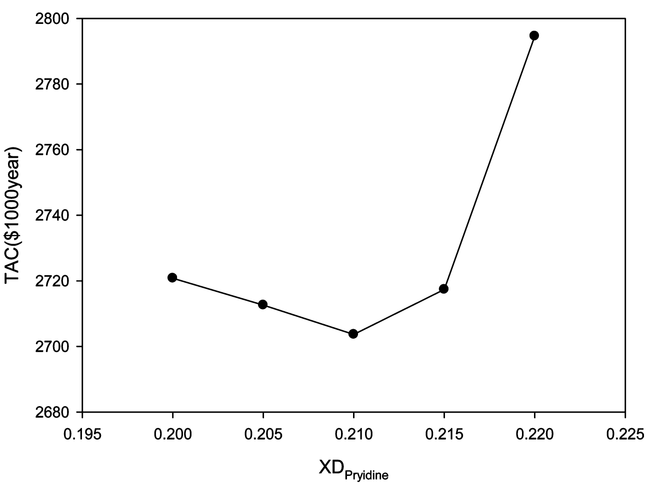

Figure 12. Summary of TAC plots at various XD2.

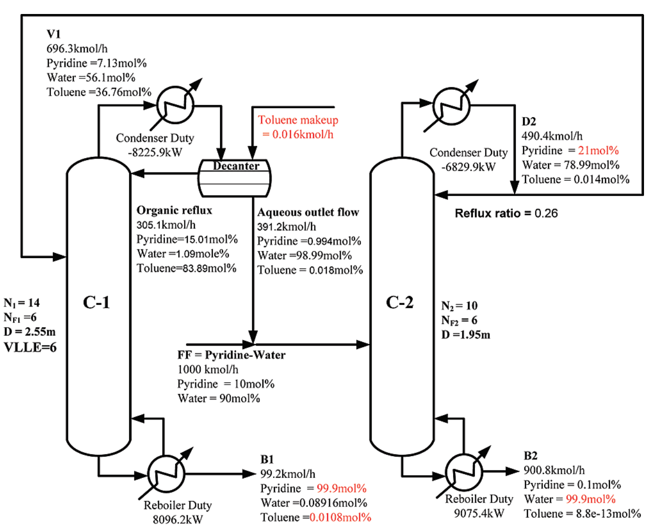

Figure 13. Detailed information of the proposed design flow sheet.

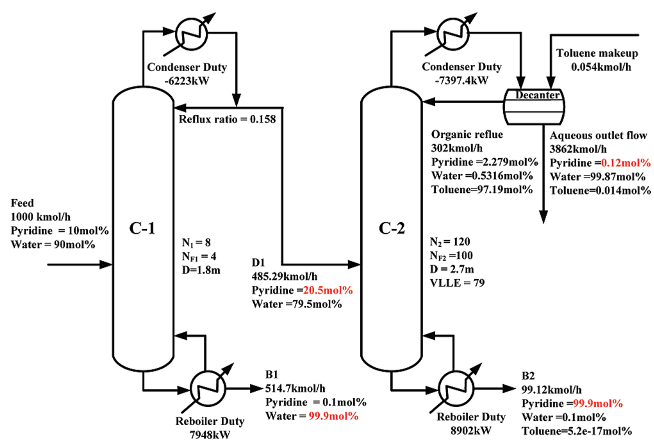

Figure 14. Detailed information of the design flow sheet with column in sequence (smaller pyridine loss, 0.12 mol %).

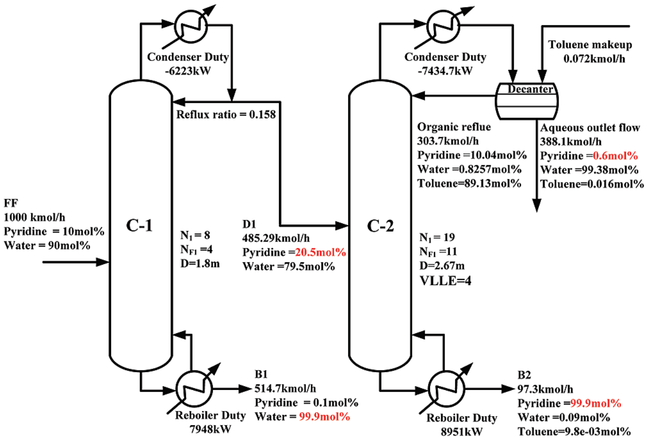

Figure 15. Detailed information of the design flow sheet with column in sequence (larger pyridine loss, 0.6 mol %).

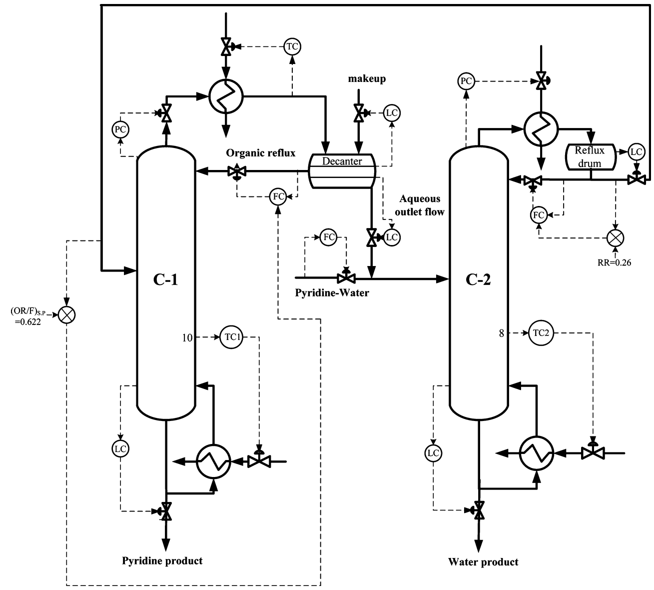

Figure 16. Proposed overall control strategy.

(0.1 mol % vs 0.6 mol %). Using the design flow sheet in this section with comparable pyridine loss, the TAC has to be increased significantly (59% more). Thus, it is obvious that the proposed design flow sheet is better than the one with two columns in sequence. In Table 2, another column shows the information with the proposed design but with larger makeup flow rate. We will explain the use of this design in the next section when we discuss the control strategy of this system.

4. Overall Control Strategy

In the following, we will investigate the proper overall control strategy for the proposed design flow sheet in Figure 13. Only tray temperature control loop(s) will be used in the overall control strategy for wider industrial applications.

The Aspen Plus steady-state simulation in the last section is exported to the dynamic simulation of Aspen Dynamics. The tray sizing option in Aspen Plus is utilized to calculate the column diameter of both columns with the assumed tray spacing of 0.6096 m. The diameters of the C-1 and the C-2 columns are calculated to be 2.55 and 1.95 m, respectively, and the weir heights of both columns are assumed to be 0.0508 m. Other equipment sizing followed the recommendation by Luyben.[32] The volumes of the column base for the C-1 column and also for the C-2 column and the reflux drum of the C-2 column are all sized to give 10 min holdup with 50% liquid level. The decanter is sized to be bigger to allow for two liquid phases to separate. The holdup time of 20 min for the decanter is used in the dynamic simulation. Pressure-driven simulation in Aspen Dynamics is used with the top stage pressures of the C-1 column set at 1.1 atm to allow for some pressure drop in the condenser and decanter. The pressure at the reflux drum of the C-2 column is set to be at atmospheric pressure. The pressure drops inside both columns are automatically calculated in Aspen Dynamics to account for liquid hydraulics and vapor traffic.

4.1. Inventory Control Loops. The inventory control loops according to our past experience are designed as follows. For the heterogeneous azeotropic column with the decanter, the aqueous-phase level is controlled by manipulating the aqueous outlet flow; the column bottom level is controlled by manipulating the bottom IPA product flow; and the column top pressure is controlled by manipulating the top vapor flow. One important loop pairing from past experience (cf. Chien et al.[14]) is that the organic phase level should be controlled by entrainer makeup flow, not by an internal recycling flow of organic reflux, so that a snowballing effect can be avoided. For the combined preconcentrator/recovery column, the reflux drum level is controlled by the distillate flow (feed to heterogeneous azeotropic column); the column bottom level is controlled by the bottom water product flow; and the column pressure is controlled by the condenser duty.

In all the closed-loop simulation runs, a P-only controller is used in all level loops for both columns. The reason for using a P-only controller is to provide maximum flow smoothing and also because maintaining a liquid level at set point value is often not necessary. $K_c = 2$ as suggested in Luyben[32] is used in most of the level loops. For the organic phase level loop to manipulate the entrainer makeup flow, $K_c = 10$ is used to simulate very fast control behavior favorable in this control loop for quickly sending more entrainer

Table 2. Comparison of Three Cases

case	proposed design flow sheet: with makeup = 0.016 kmol/h	proposed design flow sheet: with makeup = 0.09 kmol/h	design flow sheet in Figure 8: with less pyridine loss	design flow sheet in Figure 8: with larger pyridine loss
pyridine loss	0.1 mol %	0.1 mol %	0.12 mol %	0.6 mol %
C-1 column annualized equipment cost ($1000/year)	615	620	407	407
C-1 column steam cost ($1000/year)	747	756	712	712
C-1 column cooling water cost ($1000/year)	25.1	25.4	20.8	20.8
C-2 column annualized equipment cost ($1000/year)	473	476	2310	737
C-2 column steam cost ($1000/year)	819	825	813	818
C-2 column cooling water cost ($1000/year)	23.0	23.2	24.7	24.8
entrainer makeup cost ($1000/year)	0.660	3.71	2.21	2.95
total annualized capital cost ($1000/year)	1088	1096	2718	1145
total annualized operating cost ($1000/year)	1614	1633	1573	1578
TAC ($1000/year)	2702	2729	4291	2723

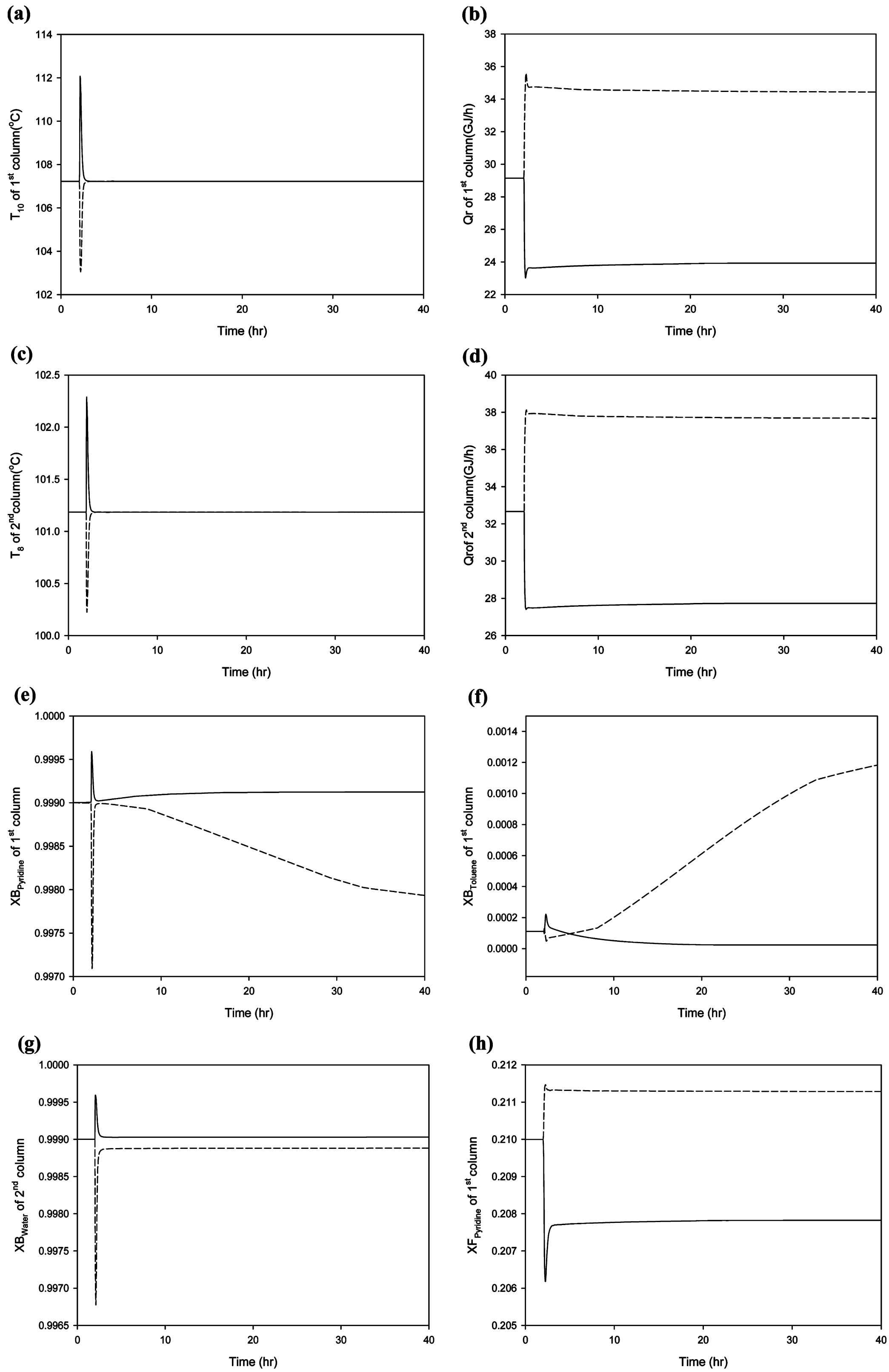

Figure 17. Closed-loop responses with ±20% feed pyridine composition changes for the design flow sheet with smaller makeup (dashed lines, +20%; solid lines, −20%).

or quickly stop feeding makeup entrainer into this system. For the top pressure control loops of both column, tight PI controller tuning parameters of $K_c = 20$ and $\tau_I = 12$ min are used.

The remaining manipulated variables for heterogeneous azeotropic column are the organic reflux flow and the reboiler duty, and the remaining manipulated variables for the preconcentrator/recovery column are the reflux flow and the reboiler duty. We will investigate the simplest overall control strategy first with only one tray temperature control loop at each column.

4.2. Tray Temperature Control Loop(s). Open-loop sensitivity analysis is used to determine the tray temperature control point for both the heterogeneous azeotropic column and the

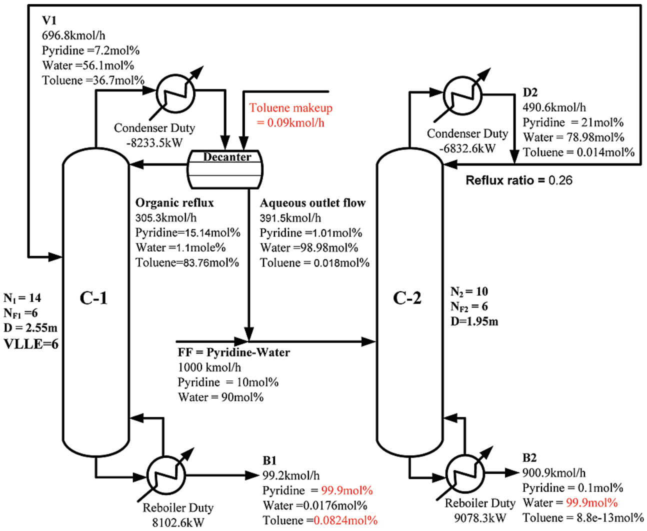

Figure 18. Detailed information of the proposed design flow sheet (larger makeup).

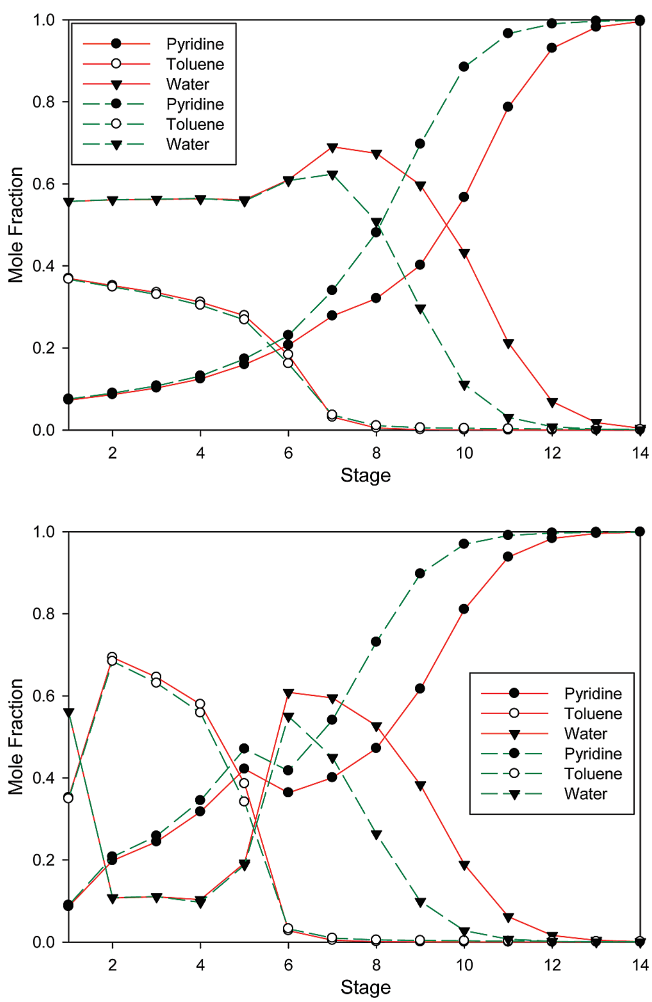

Figure 19. Vapor (top) and combined liquid (bottom) composition profiles in the heterogeneous azeotropic column: solid lines, small makeup; dashed lines, large makeup.

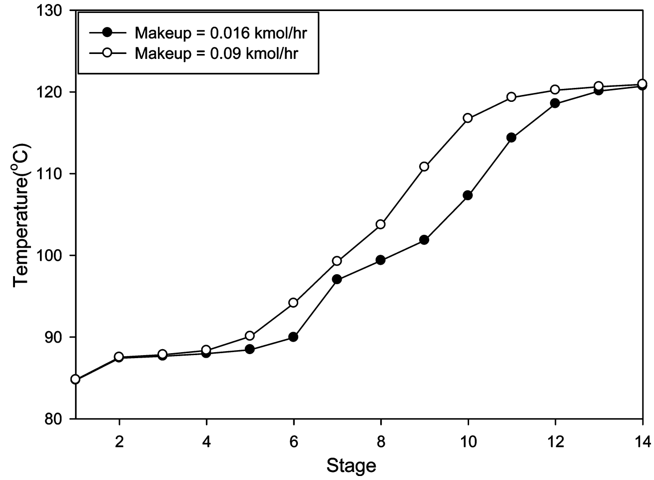

Figure 20. Temperature profile in the heterogeneous azeotropic column.

preconcentrator/recovery column. Stage no. 10 of the heterogeneous azeotropic column and stage no. 8 of the preconcentrator/recovery column are selected as the temperature control points because of high sensitivity and nearly linear behavior. PI control form is used in these two temperature control loops. In each temperature loop, an additional 1 min dead time is included for modeling the other neglected dynamics in the system. The tuning constants are determined by using the relay feedback test provided in Aspen Dynamics with Tyreus and Luyben tuning rules.[33] The iterative tuning procedure was to tune the temperature loop in the combined column first and then for the heterogeneous azeotropic column. The procedure is repeated until the tuning parameters from relay feedback test converge. The resulting PI tuning constants for tray no. 10 temperature loop of the heterogeneous azeotropic column are as follows: $K_c = 1.7$, and $\tau_I = 9.0$ min. The tuning constants for the tray no. 8 temperature loop of the preconcentrator/recovery column are as follows: $K_c = 5.3$, and $\tau_I = 6.6$ min.

The other two manipulated variables not used in the two tray temperature control loops are as follows: the organic reflux flow for the heterogeneous azeotropic column and the reflux flow for the preconcentrator/recovery column. A ratio control scheme is designed so that these two manipulated variables can be adjusted in the face of disturbances. The organic reflux flow is set to maintain a constant ratio to the feed flow of the heterogeneous azeotropic column and the reflux ratio of the preconcentrator/recovery column is also maintained. The overall proposed control strategy is summarized in Figure 16.

4.3. Simulation Results for the Design Flow Sheet in Figure 13. Two types of disturbances will be used to test the proposed control strategy. The tough disturbance of unmeasured fresh feed composition changes is considered first. Large variations of ±20% changes in the fresh feed pyridine composition are tested with the closed-loop results shown in Figure 17. From Figure 17a,c, the two tray temperatures are quickly controlled back to their set point values. For the more sensitive heterogeneous azeotropic column, its feed composition does not vary much (only change from 0.210 to 0.211 or from 0.210 to 0.208 can be seen from Figure 17h). The feed rate into this heterogeneous azeotropic column would be increased with +20% changes in the fresh feed composition because of more pyridine into the system. The feed rate into the combined column should also be increased because of more aqueous outlet flow. As a result, both reboiler duties need to be increased as can be seen from Figure 17b,d.

The dynamic response of the pyridine product stream is unacceptably slow during +20% changes in the fresh feed pyridine composition. At time = 40 h, the slippage of entrainer (toluene) from the bottom stream of the heterogeneous azeotropic column can still be observed in Figure 17f. The pyridine product purity has been dropped to 0.998 and is still falling at time = 40 h in Figure 17e. It is also noticed the initial drop of

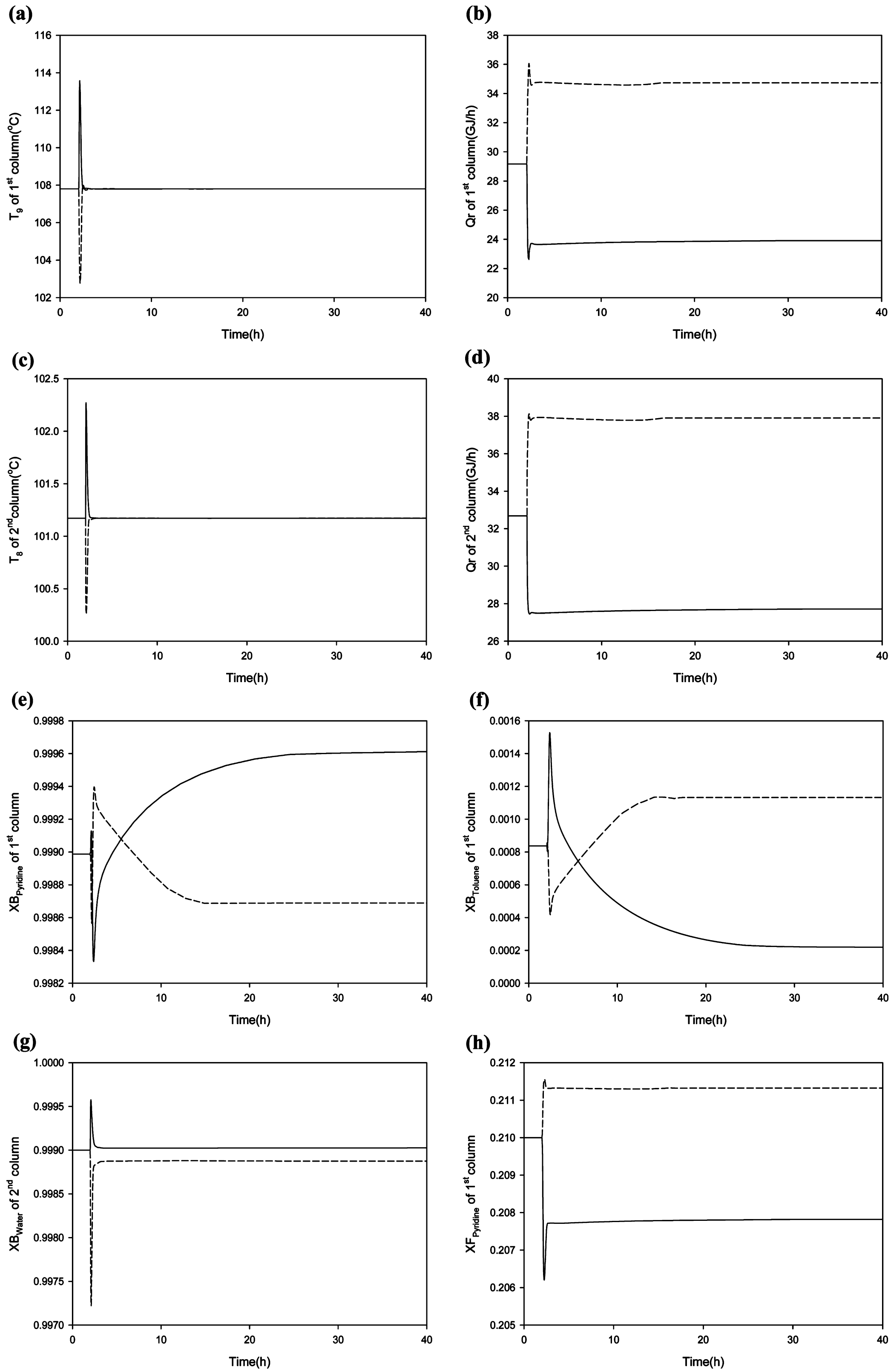

Figure 21. Closed-loop responses with ±20% feed pyridine composition changes for the design flow sheet with larger makeup (dashed lines, +20%; solid lines, −20%).

the purity to 0.997 in the pyridine product stream and then quick recovery is mainly due to the water impurity.

Because the closed-loop response was unacceptably slow, an attempt has been made to improve the dynamic response. The main reason for the slowness of the dynamic response was the slippage of entrainer from the bottom stream during transient response. The rebalancing of the entrainer inside the system by the entrainer makeup stream would take a long time. It is conjecture that if the bottom stream of the heterogeneous azeotropic column contains more entrainer at the base case operating condition, the slippage of the entrainer during transient would not have that large an impact on the overall system dynamics. The way to obtain more entrainer at the bottom stream is to slightly increase the entrainer makeup flow rate.

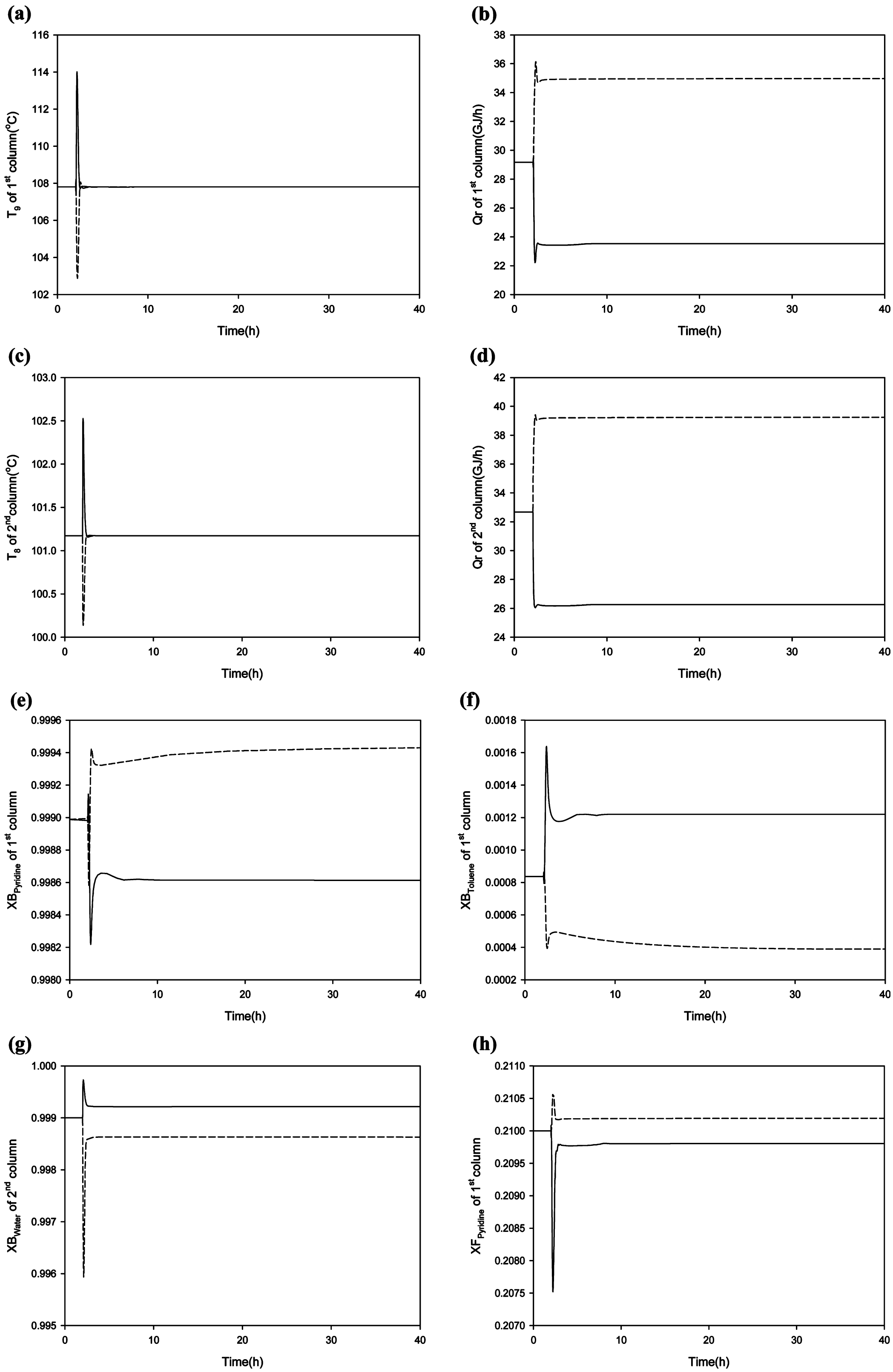

Figure 22. Closed-loop responses with ±20% feed flow rate changes for the design flow sheet with larger makeup (dashed lines, +20%; solid lines, −20%).

Figure 18 displays a new operating condition with larger makeup flow rate. It is noticed that the entrainer impurity at the bottom stream of the heterogeneous azeotropic column has been increased from 0.0108 mol % (Figure 13) to 0.0824 mol %. This new operating condition would not make a larger increase of the TAC (less than 1% increase, as can be seen in Table 2). The vapor and liquid composition profiles in the heterogeneous azeotropic column are shown in Figure 19, and the temperature profile of the two base cases are shown in Figure 20. Six stages (counting from the top stage) exhibit liquid splitting; thus, the liquid compositions for those stages are the compositions of the combined liquid. It is notice that the composition break of the pyridine product shifts more toward the middle of the column when the entrainer makeup flow rate is higher. The temperature break also moves away from the column bottoms. In the following, the closed-loop disturbance tests will be performed at this new operating condition.

4.4. Simulation Results for a Better Base Case Operating Condition as in Figure 18. The same overall control strategy was used at this new operating condition. The open-loop sensitivity tests selected stage no. 9 of the heterogeneous azeotropic column and stage no. 8 of the preconcentrator/recovery column as the controlled points. The same relay feedback tests were performed to iteratively obtain the tuning constants of the two temperature control loops.

Figure 21 displays the closed-loop responses with ±20% changes in the fresh feed pyridine composition. In comparison with the results in Figure 17, the transient response is much faster when operating at this new base case. The pyridine product purity in Figure 21e settles much faster within 20 h and with only small acceptable deviation from the base case condition of 99.9 mol %.

Another disturbance test is the throughput changes. Figure 22 displays the closed-loop responses with ±20% changes in the fresh feed flow rate. Again, the two controlled temperatures returned quickly to their set point values and with small deviations of the two product purities. Because the ±20% throughput changes are considered as known disturbances, adjustments of the set point values of the two temperature loops can be made to maintain the two product purities. Since the implementation is trivial, the closed-loop responses with the temperature adjustments are not shown here.

5. Conclusions

In this paper, design and control of a complete heterogeneous azeotropic distillation column system for pyridine dehydration has been investigated. This RCM type has not been studied before. From the design study, the flow sheet with two columns in sequence is not attractive. If tighter pyridine loss specification is enforced, an impractically tall heterogeneous azeotropic column is required. With reasonable total stages for the heterogeneous azeotropic column, the larger pyridine loss from the aqueous outlet stream is inevitable. On the contrary, the proposed design flow sheet takes into account the specification of the pyridine loss and also the limitation of a reasonably tall heterogeneous azeotropic column. The total annual cost of the proposed design flow sheet is even smaller than that of the flow sheet with larger pyridine loss.

An overall control strategy of this system is proposed with only one tray temperature control loop in each column. It is found out that a base case operating condition with slightly larger makeup flow rate would result in much faster closed-loop operation in the face of various feed disturbances. This more favorable base case can be identified by moving the temperature break more toward the middle of the heterogeneous azeotropic column and further away from the column bottoms. The slow dynamics of entrainer slippage from the bottom stream can be avoided with the clever choice of the base case operating condition.

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